Failure of uniform Sobolev estimates for concave nonlocal equations with rough kernels

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Abstract

We consider interior full-order Sobolev estimates for concave, translation-invariant nonlocal equations with even kernels comparable to that of the fractional Laplacian. Such estimates are required to be uniform over the kernel family and to depend on the forcing only through its L^p norm. We show that they fail for every strict ellipticity ratio and every finite $p > \max\{2, n/\sigma\}$, where $0 < \sigma < 2$ is the order of the equation. The counterexamples have smooth bounded solutions, smooth bounded forcing, and kernels smooth away from the origin. Independent controls on disjoint jump annuli convert small oscillations into a prescribed nonlinear average, while the linear oscillations tend to zero in L^p. Concentrated negative Riesz potentials then contradict the proposed uniform Sobolev bound.

1 Introduction

Concave integro-differential equations describe diffusion under a choice of jump law and form a natural nonlocal counterpart of concave fully nonlinear elliptic equations. Pointwise comparability of their kernels with the fractional Laplacian gives a common order and ellipticity scale. A basic regularity question is whether these assumptions also give an interior Sobolev estimate at the full operator order when the forcing is controlled in an integrable norm.

Fix $n \geq 2$, $0 < \sigma < 2$ and $0 < \lambda \leq \Lambda$. For an arbitrary family of even measurable kernels satisfying

$$\lambda|y|^{-n-\sigma} \leq K_a(y) \leq \Lambda|y|^{-n-\sigma}, \tag{1.1}$$

write

$$L_a u(x) = \mathrm{PV} \int_{\mathbb{R}^n} [u(x) - u(x+y)] K_a(y)\, \mathrm{d}y, \qquad Iu = \inf_a (-L_a u). \tag{1.2}$$

For bounded viscosity solutions of $Iu = f$ in B_1, the estimate in question is

$$\|u\|_{W^{\sigma,p}(B_{1/2})} \leq C\big(\|u\|_{L^\infty(\mathbb{R}^n)} + \|f\|_{L^p(B_1)}\big), \tag{1.3}$$

with C depending only on $n, \sigma, p, \lambda, \Lambda$, for all sufficiently large finite p. Here B_r is the ball of radius r centered at the origin. We use the Slobodeckij space when σ is noninteger and ordinary $W^{1,p}$ when $\sigma = 1$. Thus, for $0 < t < 1$,

$$[v]^p_{W^{t,p}(D)} = \int_{D\times D} \frac{|v(x) - v(y)|^p}{|x-y|^{n+tp}}\, \mathrm{d}x\, \mathrm{d}y;$$

for $1 < \sigma < 2$ the corresponding fractional seminorm is applied to ∇v at order $\sigma - 1$, together with the usual lower-order L^p norms.

A qualitative full-order Sobolev question appears as Open Question 3.2 in Fernández-Real and Ros-Oton [1, Section 3.6.2]. Here we consider the uniform quantitative estimate (1.3). A positive benchmark is Kitano's estimate for the fractional Hessian under a quadratic-angular matrix restriction on the kernels and the coefficient hypotheses in [2, Theorem 4.1]. Its Riesz-potential reduction uses that special representation. For arbitrary rough kernels, Serra's higher Hölder estimates require global Hölder control of the solution, and bounded exterior data admit counterexamples to stronger $C^{\sigma+\alpha}$ estimates [3]. Vivas obtains C^σ interior estimates for noninteger σ with bounded continuous forcing [4]. Control of increments at that order alone does not provide the scale summability or the L^p-forcing dependence in (1.3).

We show that the kernel-uniform estimate (1.3) is false. Smooth forcing does not restore it.

Theorem 1.1. *Let $n \geq 2$, $0 < \sigma < 2$, $0 < \lambda < \Lambda$, and $\max\{2, n/\sigma\} < p < \infty$. For every $C > 0$ there are a family of kernels satisfying (1.1) and functions $u, f \in C_b^\infty(\mathbb{R}^n)$ such that $Iu = f$ on $\mathbb{R}^n$ and*

$$\|u\|_{W^{\sigma,p}(B_{1/2})} > C(\|u\|_{L^\infty(\mathbb{R}^n)} + \|f\|_{L^p(B_1)}).$$

Every kernel in the family may be chosen smooth away from the origin.

In particular, with $\sigma = 2s$, no finite threshold $p_*(n, s, \lambda, \Lambda)$ can give the proposed estimate for the full class at a strict ellipticity tuple. The equal-constant case $\lambda = \Lambda$ is outside the negative assertion. The theorem concerns uniformity over the whole kernel class; individual operators may have estimates depending on additional kernel regularity.

The construction uses the freedom to choose the kernel independently on disjoint annuli of jump lengths. Minimization then produces a sum of norms of annular operators. Two prescribed cosine moments keep these norms strictly positive, so the resulting scalar forcing is smooth. A long sum of small oscillations has a nonzero nonlinear average, whereas its normalized linear oscillations tend to zero in $L^p(B_1)$ for every finite $p \geq 2$. Modulating these oscillations produces solutions with forcing tending to zero and with any prescribed smooth bounded uniform limit w satisfying $-(-\Delta)^{\sigma/2} w \geq 0$ on $\mathbb{R}^n$; see Proposition 4.1. If (1.3) held, it would bound the Sobolev norm of every such limit by its supremum norm. Negative Riesz potentials concentrated at small scales violate that bound.

Related failures of finite-p maximum estimates for kernels depending on the spatial point were recently constructed by Jiang and Xu [5]. Their use of successively smaller jump scales motivates the scale separation here. The present construction concerns translation-invariant Bellman operators and the full-order Sobolev estimate (1.3), including its global supremum term; it does not invoke their variable-coefficient theorem.

Section 2 constructs the admissible Bellman family. Section 3 supplies the scale estimates and averaging. The prescribed-limit construction is proved in Section 4, and concentration finishes the argument in Section 5.

2 Annular operators and a Bellman family

Throughout the proof, $n \geq 2$ and $0 < \sigma < 2$ are fixed. The notation C_b^∞ requires the function and each of its derivatives to be bounded. Write

$$\delta_y v(x) = v(x+y) + v(x-y) - 2v(x), \qquad \mathcal{A}v(x) = \frac{1}{2}\int_{\mathbb{R}^n} \delta_y v(x) \frac{\mathrm{d}y}{|y|^{n+\sigma}}.$$

These integrals converge absolutely for $v \in C_b^\infty(\mathbb{R}^n)$. We first construct annular operators that distinguish two frequencies. The second frequency will prevent a zero in the nonlinear term and ensure a smooth forcing.

Lemma 2.1. *There are even real functions $b_1, b_2 \in C_c^\infty(\{1 < |y| < 2\})$ such that, for*

$$m_\ell(t) = \int_{\mathbb{R}^n} (\cos(ty_1) - 1) b_\ell(y) |y|^{-n-\sigma} \, \mathrm{d}y,$$

$$(m_1(1), m_2(1)) = (1, 0), \qquad (m_1(2), m_2(2)) = (0, 1).$$

Moreover $|m_\ell(t)| \le C \min\{t^2, 1\}$ for every real t.

Proof. Choose a nonnegative radial $\chi \in C_c^\infty(\{1 < |y| < 2\})$ that is positive on an open annulus. Set $q_1(y) = \cos y_1 - 1$, $q_2(y) = \cos(2y_1) - 1$ and

$$G_{ab} = \int_{\mathbb{R}^n} \chi q_a q_b |y|^{-n-\sigma} \, \mathrm{d}y, \qquad a, b \in \{1, 2\}.$$

The matrix G is positive definite. Indeed, a zero quadratic form would force $c_1 q_1 + c_2 q_2 = 0$ on the open annulus. Since $q_2 = 2(\cos y_1 + 1) q_1$ and y_1 ranges through an interval, this forces $c_1 = c_2 = 0$. Define

$$b_\ell(y) = \chi(y) \sum_{a=1}^{2} (G^{-1})_{a\ell} q_a(y).$$

Multiplication by G gives the four moment identities. The last estimate follows by integrating $|\cos(ty_1) - 1| \le \min\{t^2 |y|^2 / 2, 2\}$ on the fixed annular support. □

Fix these functions once and for all, and put

$$M = \sup_y \sqrt{b_1(y)^2 + b_2(y)^2} > 0, \qquad \kappa = \frac{\lambda + \Lambda}{2}, \qquad \varepsilon = \frac{\Lambda - \lambda}{2M} > 0. \tag{2.1}$$

Let $R \ge 5$ and $k_0 \ge 1$ be integers and $N \ge 1$. Define

$$k_j = k_0 R^{j-1}, \qquad \mathcal{B}_j^\ell v(x) = \frac{1}{2} \int_{\mathbb{R}^n} \delta_y v(x) b_\ell(k_j y) |y|^{-n-\sigma} \, \mathrm{d}y, \qquad 1 \le j \le N,$$

and write $\mathcal{B}_j v = (\mathcal{B}_j^1 v, \mathcal{B}_j^2 v) \in \mathbb{R}^2$. For $a = (a_1, \dots, a_N) \in (\overline{B}_1^{\mathbb{R}^2})^N$, set

$$K_a(y) = \left(\kappa + \varepsilon \sum_{j=1}^{N} a_j \cdot (b_1(k_j y), b_2(k_j y)) \right) |y|^{-n-\sigma}. \tag{2.2}$$

The supports for distinct j are disjoint. Thus the coefficient in parentheses lies in $[\lambda, \Lambda]$ for every $y \ne 0$ and every a. Each K_a is even, independent of x, and smooth away from the origin. With the positive-operator convention for L_a in the Introduction, independent minimization on the disks gives the exact identity

$$I_{N,R,k_0} v = \inf_a (-L_a v) = \kappa \mathcal{A} v - \varepsilon \sum_{j=1}^{N} |\mathcal{B}_j v|. \tag{2.3}$$

The control set is compact and finite dimensional. Its minimizing index can vary with x; every member of the kernel family remains translation invariant.

3 Oscillations at separated scales

The constants in this section may depend on n, σ, b_1, b_2, but are uniform in N, $R \geq 5$, $k_0 \geq 1$ and $1 \leq j \leq N$. We will specify any additional dependence on a smooth amplitude.

Lemma 3.1. *Let $g \in C_b^\infty(\mathbb{R}^n)$ and set*

$$V_N(x) = \frac{1}{N} \sum_{i=1}^{N} k_i^{-\sigma} (\cos(k_i x_1) + \cos(2k_i x_1)), \quad v_N = gV_N, \quad z(t) = (\cos t, \cos 2t).$$

For $\alpha = \min\{\sigma, 2-\sigma\} > 0$,

$$\|\mathcal{B}_j v_N - \tfrac{g}{N} z(k_j x_1)\|_\infty \leq \frac{C\|g\|_\infty R^{-\alpha} + C_g/k_0}{N}, \tag{3.1}$$

where C_g depends only on bounded first and second derivatives of g. For $w \in C_b^\infty(\mathbb{R}^n)$ one also has

$$\|\mathcal{B}_j w\|_\infty \leq C\|D^2 w\|_\infty k_0^{\sigma-2}, \qquad \|v_N\|_\infty \leq \frac{C\|g\|_\infty k_0^{-\sigma}}{N}. \tag{3.2}$$

Proof. Scaling gives

$$\mathcal{B}_j^\ell \cos(tk_i x_1) = k_j^\sigma m_\ell(tk_i/k_j) \cos(tk_i x_1).$$

For constant g, the terms with $i = j$ give precisely $gz(k_j x_1)/N$. For $i < j$, the normalized action is bounded by $C(k_i/k_j)^{2-\sigma}$; for $i > j$ it is bounded by $C(k_j/k_i)^\sigma$. The two geometric sums are at most $CR^{-\alpha}$.

To include the amplitude, expand the product before estimating:

$$\begin{aligned}
&\delta_y(g\cos(kx_1)) - g(x)\delta_y \cos(kx_1) \\
&\quad = \cos(kx_1)\cos(ky_1)\delta_y g(x) - \sin(kx_1)\sin(ky_1)(g(x+y) - g(x-y)).
\end{aligned}$$

On the jth annulus, the error from the ith normalized mode is bounded by

$$Ck_i^{-\sigma}\left(k_j^{\sigma-2}\|D^2 g\|_\infty + k_j^{\sigma-1}\min\{1, k_i/k_j\}\|\nabla g\|_\infty\right). \tag{3.3}$$

The sum of the first terms is at most $Ck_0^{-2}\|D^2 g\|_\infty$. For the coarser-frequency part of the second term, use

$$k_j^{\sigma-2} \sum_{i \leq j} k_i^{1-\sigma} \leq C/k_0.$$

For $\sigma < 1$ the sum is controlled by its largest term, and for $\sigma > 1$ by its first term. When $\sigma = 1$, the bound is $j/k_j \leq C/k_0$. The finer-frequency sum is at most C/k_j. Dividing the summed errors by N proves (3.1). Taylor's formula on the annular support proves the first estimate in (3.2); the second is the geometric sum defining V_N. □

Lemma 3.2. *Put*

$$c_\sigma = \int_{\mathbb{R}^n} (1 - \cos y_1)|y|^{-n-\sigma}\, dy > 0, \qquad T(t) = \cos t + 2^\sigma \cos 2t.$$

For each fixed $g \in C_b^\infty(\mathbb{R}^n)$ there is a function $\eta_g(K) \to 0$ as $K \to \infty$ such that

$$\mathcal{A}(gV_N) = -c_\sigma g(x)\frac{1}{N}\sum_{j=1}^{N} T(k_j x_1) + e_N(x), \qquad \|e_N\|_\infty \leq \eta_g(k_0). \tag{3.4}$$

Proof. The defining integral for c_σ is finite because $0 < \sigma < 2$. Scaling and evenness give $\mathcal{A}\cos(kx_1) = -c_\sigma k^\sigma \cos(kx_1)$. We claim that

$$k^{-\sigma}\|\mathcal{A}(g\cos(kx_1)) - g\mathcal{A}\cos(kx_1)\|_\infty \longrightarrow 0.$$

In the integral for this difference set $y = z/k$. The product expansion in the previous proof gives the integrable majorant $C_g \min\{|z|^2, 1\}|z|^{-n-\sigma}$, uniformly in x and $k \geq 1$. For each fixed z, uniform continuity of g makes the supremum in x of the integrand's coefficient tend to zero. The supremum is measurable, since it can be taken over rational x. Dominated convergence proves the claim. Apply it also to the second harmonic. A tail supremum of these two commutator errors defines η_g and proves (3.4). □

Lemma 3.3. *Let F be a smooth 2π-periodic function with mean $\overline{F}$, and fix an integer $R \geq 5$. For every finite $p \geq 2$,*

$$\left\| \frac{1}{N}\sum_{j=1}^{N} (F(k_0 R^{j-1}x_1) - \overline{F}) \right\|_{L^p(B_1)} \longrightarrow 0, \tag{3.5}$$

uniformly over integers $k_0 \geq 1$.

Proof. First take a mean-zero trigonometric polynomial P of degree D. Fourier orthogonality on a period shows that scales i, j can overlap only if $mR^i = \ell R^j$ for nonzero integers m, ℓ of magnitude at most D. Consequently $|i - j| \leq \log_R D$ and

$$\left\| \frac{1}{N}\sum_{j=1}^{N} P(R^{j-1}t) \right\|^2_{L^2(0,2\pi)} \leq C_{P,R}/N.$$

Uniform trigonometric approximation extends this convergence to $F - \overline{F}$. The averages are uniformly bounded, so their L^p norms tend to zero for each finite $p \geq 2$, by $\|S\|_p^p \leq \|S\|_\infty^{p-2}\|S\|_2^2$. For any periodic S, covering $[-k_0, k_0]$ by full periods and at most two additional periods gives

$$\int_{-1}^{1} |S(k_0 t)|^p \, \mathrm{d}t \leq C \int_0^{2\pi} |S(t)|^p \, \mathrm{d}t,$$

with C independent of $k_0 \geq 1$. Integration of the bounded cross sections of B_1 now proves (3.5). □

The particular nonlinear average that we need is

$$H(t) = |z(t)| = \sqrt{\cos^2 t + \cos^2(2t)}, \qquad \mu = \frac{1}{2\pi}\int_0^{2\pi} H(t)\,\mathrm{d}t > 0. \tag{3.6}$$

It is smooth and has the lower bound

$$H(t)^2 = 4\cos^4 t - 3\cos^2 t + 1 \geq 7/16. \tag{3.7}$$

This lower bound is the reason for using two separated moments rather than one absolute cosine.

4 Small smooth forcing and a prescribed limit

The previous lemmas let us choose a smooth amplitude so that the nonlinear oscillations cancel the generator of a prescribed background.

Proposition 4.1. *Let $w \in C_b^\infty(\mathbb{R}^n)$ satisfy $\mathcal{A}w \geq 0$ on $\mathbb{R}^n$, and let $2 \leq p < \infty$. There are operators I_m of the form (2.3), with the same $n, \sigma, \lambda, \Lambda$, and functions $u_m, f_m \in C_b^\infty(\mathbb{R}^n)$ such that*

$$I_m u_m = f_m \quad \text{on } \mathbb{R}^n, \qquad \|f_m\|_{L^p(B_1)} \to 0, \qquad \|u_m - w\|_{L^\infty(\mathbb{R}^n)} \to 0. \tag{4.1}$$

The choices can also ensure uniform convergence of first derivatives when $\sigma > 1$, and convergence in ordinary $W^{1,p}(B_1)$ when $\sigma = 1$.

Proof. Put $q = \mathcal{A}w \geq 0$. Differentiation under the integral, using Taylor's formula near zero and bounded derivatives at infinity, gives $q \in C_b^\infty(\mathbb{R}^n)$. Given $\delta > 0$, fix the positive amplitude

$$g(x) = \frac{\kappa q(x) + \delta}{\varepsilon \mu}, \qquad g_{\min} = \frac{\delta}{\varepsilon \mu} > 0. \tag{4.2}$$

Set $u_N = w + gV_N$ and $f_N = I_{N,R,k_0} u_N$. From (3.1)–(3.2),

$$\begin{aligned} \sum_{j=1}^{N} \|\mathcal{B}_j u_N - \tfrac{g}{N} z(k_j x_1)\|_\infty &\leq CN \|D^2 w\|_\infty k_0^{\sigma-2} + C\|g\|_\infty R^{-\alpha} + C_g/k_0 \\ &=: \mathcal{E}. \end{aligned} \tag{4.3}$$

Using the 1-Lipschitz property of the Euclidean norm and $\kappa q - \varepsilon \mu g = -\delta$, we obtain

$$\begin{aligned} f_N = &-\delta - \kappa c_\sigma g(x) \frac{1}{N} \sum_j T(k_j x_1) \\ &- \varepsilon g(x) \left(\frac{1}{N} \sum_j H(k_j x_1) - \mu \right) + r_N, \\ \|r_N\|_\infty &\leq \kappa \eta_g(k_0) + \varepsilon \mathcal{E}. \end{aligned} \tag{4.4}$$

The order of choices matters. First choose the integer $R \geq 5$ so that $C\|g\|_\infty R^{-\alpha}$ is sufficiently small compared with δ and $g_{\min}\sqrt{7}/4$. Keep R and g fixed and use Lemma 3.3 to choose N so that both oscillatory terms in (4.4) have $L^p(B_1)$ norm at most δ. We may require N to be arbitrarily large. Finally choose the integer k_0 sufficiently large that $\kappa \eta_g(k_0)$, $CN\|D^2 w\|_\infty k_0^{\sigma-2}$, C_g/k_0 and $\|gV_N\|_\infty$ are as small as needed. These requirements are compatible because $\sigma < 2$ and the averaging estimate is uniform in k_0.

We also impose a pointwise condition to make the forcing smooth. The individual error satisfies

$$\|\mathcal{B}_j u_N - \tfrac{g}{N} z(k_j x_1)\|_\infty \leq C\|D^2 w\|_\infty k_0^{\sigma-2} + \frac{C\|g\|_\infty R^{-\alpha} + C_g/k_0}{N}.$$

In the first choice take the R term below $g_{\min}\sqrt{7}/16$; in the last choice take the sum of the other two terms, after multiplication by N, below the same number. Then

$$|\mathcal{B}_j u_N(x)| \geq \frac{g_{\min}\sqrt{7}}{8N} > 0 \qquad (x \in \mathbb{R}^n,\ 1 \leq j \leq N).$$

Thus each $|\mathcal{B}_j u_N|$ is smooth with bounded derivatives for this finite construction. The same holds for $\mathcal{A}u_N$, so $f_N \in C_b^\infty(\mathbb{R}^n)$ exactly. Sending $\delta \downarrow 0$ with these successive choices, and requiring $N, k_0 \to \infty$, proves (4.1).

It remains to record the derivative convergence. The amplitudes g_δ have uniformly bounded derivatives as $0 < \delta \le 1$. For $\sigma > 1$, differentiating the finite sum gives

$$\|\nabla(gV_N)\|_\infty \le \frac{C}{N}\left(\|\nabla g\|_\infty k_0^{-\sigma} + \|g\|_\infty k_0^{1-\sigma}\right) \to 0.$$

For $\sigma = 1$, transverse derivatives tend to zero uniformly and

$$\partial_1(gV_N) = (\partial_1 g)V_N - \frac{g}{N}\sum_j (\sin(k_j x_1) + 2\sin(2k_j x_1)).$$

The sine average has pairwise disjoint frequencies for $R \ge 5$. Its periodic L^2 norm is $O(N^{-1/2})$ and its supremum is at most 3. It therefore tends to zero in each finite $L^p(B_1)$, uniformly under the subsequent choice of k_0, by the period-covering argument above. This proves the last assertion. □

Every equality in Proposition 4.1 is classical. It is also a viscosity equality in the original convention: replacing u near an upper contact by a touching test function increases every $-L_a$, and at a lower contact decreases them, because every kernel is nonnegative.

5 Concentration and failure of the estimate

We finish by applying the prescribed-limit construction to concentrated negative potentials. The required sign and regularity can be checked directly.

Lemma 5.1. *There is a nonzero nonconstant $V \in C_b^\infty(\mathbb{R}^n)$ that tends to zero at infinity and satisfies $\mathcal{A}V = -c_\sigma\psi$ for some nonzero $0 \le \psi \in C_c^\infty(\mathbb{R}^n)$, where $c_\sigma > 0$ is the constant defined in Lemma 3.2.*

Proof. Here $n > \sigma$. With $h_t(x) = (4\pi t)^{-n/2}e^{-|x|^2/(4t)}$, set

$$R_\sigma(x) = \frac{1}{\Gamma(\sigma/2)}\int_0^\infty t^{\sigma/2-1}h_t(x)\,\mathrm{d}t = \frac{\Gamma((n-\sigma)/2)}{2^\sigma \pi^{n/2}\Gamma(\sigma/2)}|x|^{\sigma-n} \quad (x \ne 0),$$

and $V = R_\sigma * \psi$, where ψ is any such nonnegative smooth function. The kernel is locally integrable and positive. Moving derivatives onto ψ shows that $V \in C_b^\infty(\mathbb{R}^n)$; compact support gives $V(x) = O(|x|^{\sigma-n})$ at infinity. Thus V is nonzero and nonconstant.

For the sign, use the Fourier convention $\widehat{\psi}(\xi) = \int e^{-ix\cdot\xi}\psi(x)\,\mathrm{d}x$. The Gaussian formula $\widehat{h}_t(\xi) = e^{-t|\xi|^2}$ and the positive heat representation give

$$V(x) = (2\pi)^{-n}\int_{\mathbb{R}^n} e^{ix\cdot\xi}|\xi|^{-\sigma}\widehat{\psi}(\xi)\,\mathrm{d}\xi.$$

All interchanges are absolute: the integral over (t,ξ) is bounded by $\Gamma(\sigma/2)\int|\xi|^{-\sigma}|\widehat{\psi}(\xi)|\,\mathrm{d}\xi < \infty$, using $n > \sigma$ near zero and rapid decay at infinity. Isotropy gives $\mathcal{A}e^{ix\cdot\xi} = -c_\sigma|\xi|^\sigma e^{ix\cdot\xi}$. The absolute double integral for applying $\mathcal{A}$ to this representation is bounded by $c_\sigma\int|\widehat{\psi}| < \infty$. Hence Fubini and Fourier inversion give $\mathcal{A}V = -c_\sigma\psi$ pointwise. This also verifies the normalization directly; compare [6, Definition 2.12]. □

Proof of Theorem 1.1. Suppose the proposed estimate holds with a constant C uniform over the kernel families. Fix any $w \in C_b^\infty(\mathbb{R}^n)$ with $\mathcal{A}w \geq 0$ and apply it to Proposition 4.1. If $0 < \sigma < 1$, Fatou's lemma applied to the Slobodeckij difference-quotient density gives a lower bound for the limit norm. If $1 < \sigma < 2$, use the uniform gradient convergence and apply Fatou at order $\sigma - 1$ to the gradients. At $\sigma = 1$, use the asserted ordinary $W^{1,p}$ convergence. In all cases,

$$\|w\|_{W^{\sigma,p}(B_{1/2})} \leq C\|w\|_{L^\infty(\mathbb{R}^n)}. \tag{5.1}$$

Take V from Lemma 5.1 and $w_r(x) = -V(x/r)$. Then $\|w_r\|_\infty = \|V\|_\infty$, while $\mathcal{A}w_r = c_\sigma r^{-\sigma}\psi(x/r) \geq 0$. For $0 < \sigma < 1$, choose a fixed ball on which V has a positive Slobodeckij seminorm. Restricting the norm of w_r to its rescaled ball inside $B_{1/2}$ and changing variables gives

$$[w_r]_{W^{\sigma,p}(B_{1/2})} \geq cr^{n/p-\sigma}.$$

For $1 < \sigma < 2$, the same change of variables for the gradient at order $\sigma - 1$ gives the identical power. The corresponding seminorm is positive on some ball, since a nonzero decaying V cannot be affine. For $\sigma = 1$, choose a ball where ∇V is nonzero and obtain $\|\nabla w_r\|_{L^p(B_{1/2})} \geq cr^{n/p-1}$. Since $p > n/\sigma$, each lower bound tends to infinity as $r \downarrow 0$, contradicting (5.1). □

References

[1] X. Fernández-Real and X. Ros-Oton, *Integro-Differential Elliptic Equations*, Progress in Mathematics 350, Birkhäuser, 2024. doi:10.1007/978-3-031-54242-8. Author manuscript: arXiv:2411.12455v1.

[2] S. Kitano, *$W^{\sigma,p}$ a priori estimates for fully nonlinear integro-differential equations*, 2022. arXiv:2207.06728v1.

[3] J. Serra, *$C^{\sigma+\alpha}$ regularity for concave nonlocal fully nonlinear elliptic equations with rough kernels*, 2015. arXiv:1405.0930v3.

[4] H. Vivas, *C^{2s} regularity for fully nonlinear nonlocal equations with bounded right hand side*, 2019. arXiv:1907.02455v3.

[5] Y. Jiang and L. Xu, *A counterexample to integral Aleksandrov–Bakelman–Pucci estimates for a nonlocal operator*, 2026. arXiv:2610.16162v1.

[6] M. Kwaśnicki, *Ten equivalent definitions of the fractional Laplace operator*, 2015. arXiv:1507.07356v2.

AI-use statement

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